

FUNDAMENTAL ORGANIC ALA MECH



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Fundamental Organic ala Mech

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CHAPTER OVERVIEW

1: Atomic Structure

[1.1: Atomic Structure - The Nucleus](#)

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1.1: Atomic Structure - The Nucleus

As you might remember from your general chemistry course, an atom consists of a dense, positively charged nucleus surrounded at a relatively large distance by negatively charged electrons (Figure 1.1.1). The nucleus consists of subatomic particles called *neutrons*, which are electrically neutral, and *protons*, which are positively charged. Because an atom is neutral overall, the number of positive protons in the nucleus and the number of negative electrons surrounding the nucleus are the same.

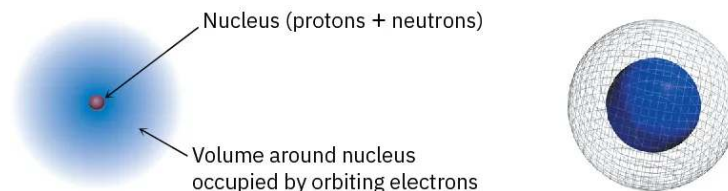


Figure 1.1.1: A schematic view of an atom. The dense, positively charged nucleus contains most of the atom's mass and is surrounded by negatively charged electrons. The three-dimensional view on the right shows calculated electron-density surfaces. Electron density increases steadily toward the nucleus and is 40 times greater at the **blue solid surface** than at the **gray mesh surface**.

Although extremely small—about 10^{-14} to 10^{-15} meter (m) in diameter—the nucleus nevertheless contains essentially all the mass of the atom. Electrons have negligible mass and circulate around the nucleus at a distance of approximately 10^{-10} m. Thus, the diameter of a typical atom is about 2×10^{-10} m, or 200 picometers (pm), where $1 \text{ pm} = 10^{-12} \text{ m}$. To give you an idea of how small this is, a thin pencil line is about 3 million carbon atoms wide. Although most chemists throughout the world use the International System (SI) of units and describe small distances in picometers, many organic chemists and biochemists in the United States still use the unit angstrom ($\text{\AA}$) to express atomic distances, where $1 \text{ \AA} = 100 \text{ pm} = 10^{-10} \text{ m}$. As you probably did in your general chemistry course, however, we'll stay with SI units in this book.

A specific atom is described by its atomic number (Z), which gives the number of protons (or electrons) it contains, and its mass number (A), which gives the total number of protons and neutrons in its nucleus. All the atoms of a given element have the same atomic number: 1 for hydrogen, 6 for carbon, 15 for phosphorus, and so on; but they can have different mass numbers depending on how many neutrons they contain. Atoms with the same atomic number but different mass numbers are called isotopes. The element carbon, for instance, has three isotopes that occur naturally, with mass numbers of 12, 13, and 14. Carbon-12 has a natural abundance of 98.89%, carbon-13 has a natural abundance of 1.11%, and carbon-14 has only a negligible natural abundance.

The weighted-average of an element's naturally occurring isotopes is called **atomic weight** and is given in unified atomic mass units (u) or daltons (Da) where 1 u or 1 Da is defined as one twelfth the mass of one atom of carbon-12. Thus, the atomic weight is 1.008 u for hydrogen, 12.011 u for carbon, 30.974 u for phosphorus, and so on. Atomic weights of all elements are given in the periodic table in Appendix D.

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1.2: Atomic Structure - Orbitals

How are the electrons distributed in an atom? You might recall from your general chemistry course that, according to the *quantum mechanical model*, the behavior of a specific electron in an atom can be described by a mathematical expression called a *wave equation*—the same type of expression used to describe the motion of waves in a fluid. The solution to a wave equation is called a *wave function*, or orbital, and is denoted by the lowercase Greek letter psi (ψ).

When the square of the wave function, ψ^2 , is plotted in three-dimensional space, an orbital describes the volume of space around a nucleus that an electron is most likely to occupy. You might therefore think of an orbital as looking like a photograph of the electron taken at a slow shutter speed. In such a photo, the orbital would appear as a blurry cloud, indicating the region of space where the electron has been. This electron cloud doesn't have a sharp boundary, but for practical purposes we can set the limits by saying that an orbital represents the space where an electron spends 90% to 95% of its time.

What do orbitals look like? There are four different kinds of orbitals, denoted *s*, *p*, *d*, and *f*, each with a different shape. Of the four, we'll be concerned primarily with *s* and *p* orbitals because these are the most common in organic and biological chemistry. An *s* orbital has a spherical shape, with the nucleus at its center; a *p* orbital has a dumbbell shape with two parts, or *lobes*; and four of the five *d* orbitals have a cloverleaf shape with four lobes, as shown in Figure 1.2.1. The fifth *d* orbital is shaped like an elongated dumbbell with a doughnut around its middle.

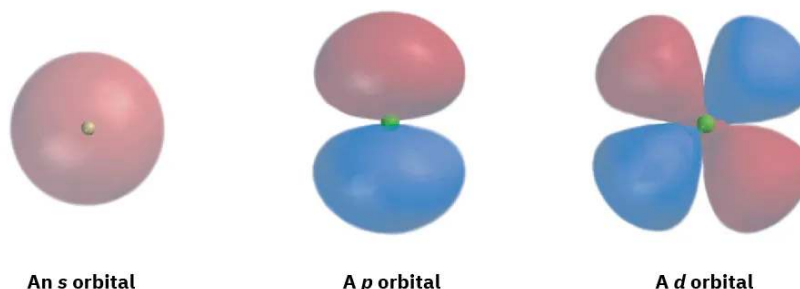


Figure 1.2.1: Representations of *s*, *p*, and *d* orbitals. An *s* orbital is spherical, a *p* orbital is dumbbell-shaped, and four of the five *d* orbitals are cloverleaf-shaped. Different lobes of *p* orbitals are often drawn for convenience as teardrops, but their actual shape is more like that of a doorknob, as indicated.

The orbitals in an atom are organized into different layers around the nucleus called electron shells, which are centered around the nucleus and have successively larger size and energy. Different shells contain different numbers and kinds of orbitals, and each orbital within a shell can be occupied by two electrons. The first shell contains only a single *s* orbital, denoted *1s*, and thus holds only 2 electrons. The second shell contains one *2s* orbital and three *2p* orbitals and thus holds a total of 8 electrons. The third shell contains a *3s* orbital, three *3p* orbitals, and five *3d* orbitals, for a total capacity of 18 electrons. These orbital groupings and their energy levels are shown in Figure 1.2.2.

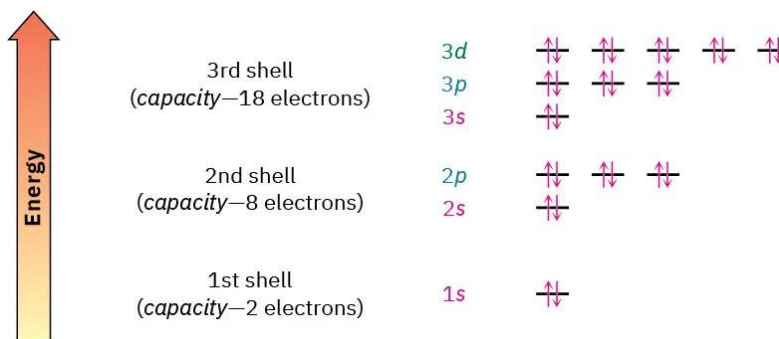


Figure 1.2.2: Energy levels of electrons in an atom. The first shell holds a maximum of 2 electrons in one *1s*

orbital; the second shell holds a maximum of 8 electrons in one *2s* and three *2p* orbitals; the third shell holds a maximum of 18 electrons in one *3s*, three *3p*, and *3d* orbitals; and so on. The two electrons in each orbital are represented by five up and down arrows, $\uparrow\downarrow$. Although not shown, the energy level of the *4s* orbital falls between *3p* and *3d*.

The three different *p* orbitals within a given shell are oriented in space along mutually perpendicular directions, denoted p_x , p_y , and p_z . As shown in Figure 1.2.3, the two lobes of each *p* orbital are separated by a region of zero electron density called a node.

Furthermore, the two orbital regions separated by the node have different algebraic signs, + and -, in the wave function, as represented by the different colors in Figure 1.2.1 and Figure 1.2.3. As we'll see in **Section 1.12**, these algebraic signs for different orbital lobes have important consequences with respect to chemical bonding and chemical reactivity.

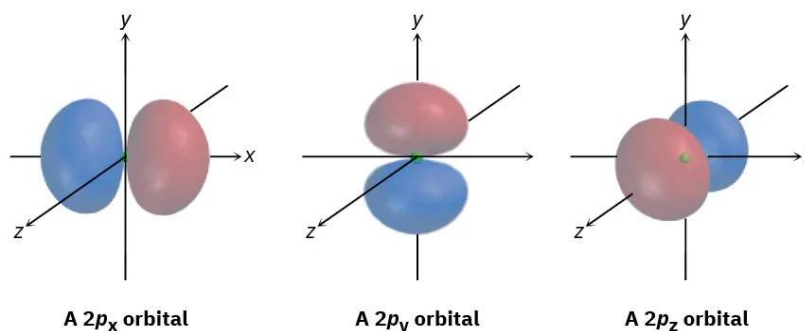


Figure 1.2.3: Shapes of the $2p$ orbitals. Each of the three mutually perpendicular, dumbbell-shaped orbitals has two lobes separated by a node. The two lobes have different algebraic signs in the corresponding wave function, as indicated by the different colors.

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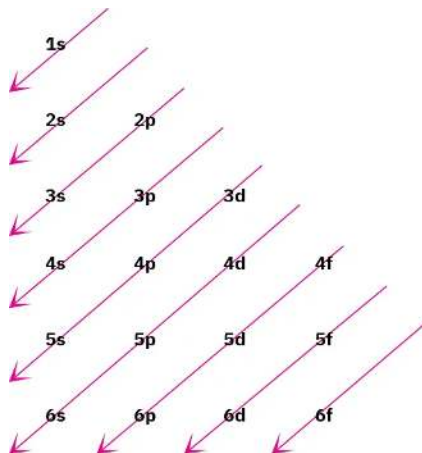
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1.3: Atomic Structure - Electron Configurations

The lowest-energy arrangement, or ground-state electron configuration, of an atom is a list of the orbitals occupied by its electrons. We can predict this arrangement by following three rules.

RULE 1

The lowest-energy orbitals fill up first, $1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s \rightarrow 3d$, according to the following graphic, a statement called the *Aufbau principle*. Note that the $4s$ orbital lies between the $3p$ and $3d$ orbitals in energy.



RULE 2

Electrons act in some ways as if they were spinning around an axis, somewhat as the earth spins. This spin can have two orientations, denoted as up ($\uparrow$) and down ($\downarrow$). Only two electrons can occupy an orbital, and they must have opposite spins, a statement called the *Pauli exclusion principle*.

RULE 3

If two or more empty orbitals of equal energy are available, one electron occupies each with spins parallel until all orbitals are half-full, a statement called *Hund's rule*.

Some examples of how these rules apply are shown in Table 1.3.1. Hydrogen, for instance, has only one electron, which must occupy the lowest-energy orbital. Thus, hydrogen has a $1s$ ground-state configuration. Carbon has six electrons and the ground-state configuration $1s^2 2s^2 2p_x^1 2p_y^1$, and so forth. Note that a superscript is used to represent the number of electrons in a particular orbital.

Table 1.3.1: Ground-State Electron Configurations of Some Elements

Element	Atomic number	Configuration
Hydrogen	1	$1s \uparrow$
Carbon	6	$1s \uparrow\downarrow$ $2s \uparrow\downarrow$ $2p \uparrow \uparrow \text{—}$
Phosphorus	15	$1s \uparrow\downarrow$ $2s \uparrow\downarrow$ $2p \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow$ $3s \uparrow\downarrow$ $3p \uparrow \uparrow \uparrow$

? Exercise 1.3.1

What is the ground-state electron configuration of each of the following elements:

- a. Oxygen
- b. Nitrogen
- c. Sulfur

Answer

- a. $1s^2 2s^2 2p^4$
- b. $1s^2 2s^2 2p^3$
- c. $1s^2 2s^2 2p^6 3s^2 3p^4$

? Exercise 1.3.2

How many electrons does each of the following biological trace elements have in its outermost electron shell?

- a. Magnesium
- b. Cobalt
- c. Selenium

Answer

- a. 2
- b. 2
- c. 6

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CHAPTER OVERVIEW

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- 2.1: Development of Chemical Bonding Theory
- 2.2: Describing Chemical Bonds - Valence Bond Theory
- 2.3: Drawing Chemical Structures
- 2.4: sp^3 Hybrid Orbitals and the Structure of Methane
- 2.5: sp^3 Hybrid Orbitals and the Structure of Ethane
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2.1: Development of Chemical Bonding Theory

By the mid-1800s, the new science of chemistry was developing rapidly, especially in Europe, and chemists had begun to probe the forces holding compounds together. In 1858, the German chemist August Kekulé and the Scottish chemist Archibald Couper independently proposed that, in all organic compounds, carbon is *tetravalent*—it always forms four bonds when it joins other elements to form stable compounds. Furthermore, said Kekulé, carbon atoms can bond to one another to form extended chains of linked atoms. In 1865, Kekulé provided another major advance when he suggested that carbon chains can double back on themselves to form *rings* of atoms.

Although Kekulé and Couper were correct in describing the tetravalent nature of carbon, chemistry was still viewed in a two-dimensional way until 1874. In that year, the Dutch chemist Jacobus van 't Hoff and French chemist Joseph Le Bel added a third dimension to our ideas about organic compounds when they proposed that the four bonds of carbon are not oriented randomly but have specific spatial directions. Van't Hoff went even further and suggested that the four atoms to which carbon is bonded sit at the corners of a regular tetrahedron, with carbon in the center.

A representation of a tetrahedral carbon atom is shown in Figure 2.1.1. Note the conventions used to show three-dimensionality: solid lines represent bonds in the plane of the page, the heavy wedged line represents a bond coming out of the page toward the viewer, and the dashed line represents a bond receding back behind the page, away from the viewer. Get used to them; these representations will be used throughout the text.

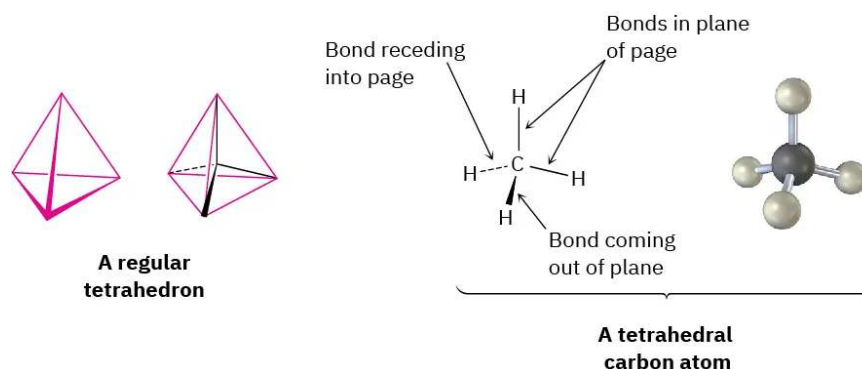


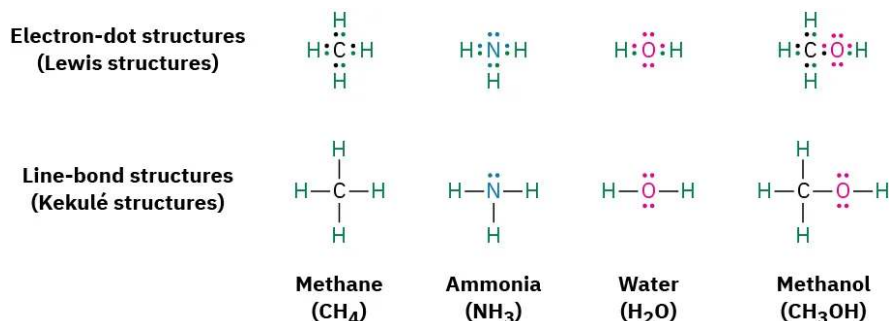
Figure 2.1.1: A representation of van't Hoff's tetrahedral carbon atom. The solid lines represent bonds in the plane of the paper, the heavy wedged line represents a bond coming out of the plane of the page toward the viewer, and the dashed line represents a bond going back behind the plane of the page away from the viewer.

Why, though, do atoms bond together, and how can chemical bonds be described electronically? The *why* question is relatively easy to answer: atoms bond together because the compound that results is more stable and lower in energy than the separate atoms. Energy—usually as heat—is always released and flows out of the chemical system when a bond forms. Conversely, energy is added to the chemical system when a bond breaks. Making bonds always releases energy, and breaking bonds always absorbs energy. The *how* question is more difficult. To answer it, we need to know more about the electronic properties of atoms.

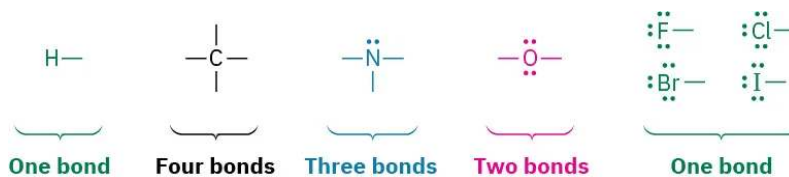
We know through observation that eight electrons (an electron *octet*) in an atom's outermost shell, or valence shell, impart special stability to the noble-gas elements in group 8A of the periodic table: Ne (2 + 8); Ar (2 + 8 + 8); Kr (2 + 8 + 18 + 8). We also know that the chemistry of the main-group elements on the left and right sides of the periodic table is governed by their tendency to take on the electron configuration of the nearest noble gas. The alkali metals such as sodium in group 1A, for example, achieve a noble-gas configuration by losing the single *s* electron from their valence shell to form a cation, while the halogens such as chlorine in group 7A achieve a noble-gas configuration by gaining a *p* electron to fill their valence shell and form an anion. The resultant ions are held together in compounds like $\text{Na}^+ \text{Cl}^-$ by the electrical attraction of unlike charges that we call an ionic bond.

But how do elements closer to the middle of the periodic table form bonds? Look at methane, CH_4 , the main constituent of natural gas, for example. The bonding in methane is not ionic because it would take too much energy for carbon ($1s^2 2s^2 2p^2$) either to gain or lose four electrons to achieve a noble-gas configuration. Instead, carbon bonds to other atoms, not by gaining or losing electrons, but by *sharing* them. Such a shared-electron bond, first proposed in 1916 by the American chemist G. N. Lewis, is called a covalent bond. The neutral collection of atoms held together by covalent bonds is called a molecule. Ionic compounds such as sodium chloride, however, are not called molecules.

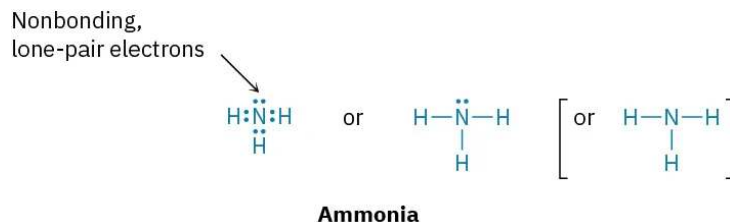
A simple way of indicating the covalent bonds in molecules is to use what are called Lewis structures, or electron-dot structures, in which the valence-shell electrons of an atom are represented as dots. Thus, hydrogen has one dot representing its $1s$ electron, carbon has four dots ($2s^2 2p^2$), oxygen has six dots ($2s^2 2p^4$), and so on. A stable molecule results whenever a noble-gas configuration of eight dots (an octet) is achieved for all main-group atoms or two dots for hydrogen. Even simpler than Lewis structures is the use of Kekulé structures, or line-bond structures, in which the two-electron covalent bonds are indicated as lines drawn between atoms.



The number of covalent bonds an atom forms depends on how many additional valence electrons it needs to reach a noble-gas configuration. Hydrogen has one valence electron ($1s$) and needs only one more to reach the helium configuration ($1s^2$), so it forms one bond. Carbon has four valence electrons ($2s^2 2p^2$) and needs four more to reach the neon configuration ($2s^2 2p^6$), so it forms four bonds. Nitrogen has five valence electrons ($2s^2 2p^3$), needs three more, and forms three bonds; oxygen has six valence electrons ($2s^2 2p^4$), needs two more, and forms two bonds; and the halogens have seven valence electrons, need one more, and form one bond.



Valence electrons that are not used for bonding remain as dots in structures and are called lone-pair electrons, or nonbonding electrons. The nitrogen atom in ammonia, NH₃, for instance, shares six valence electrons in three covalent bonds and has its remaining two valence electrons as two dots in a nonbonding lone pair. As a time-saving shorthand, nonbonding electrons are often omitted when drawing line-bond structures, but you still have to keep them in mind since they're often crucial in chemical reactions.



✓ Worked Example 2.1.1: Predicting the Number of Bonds Formed by Atoms in Molecules

How many hydrogen atoms does phosphorus bond to in forming phosphine, PH₃?

Strategy

Identify the periodic group of phosphorus, and find from that how many electrons (bonds) are needed to make an octet.

Solution

Phosphorus is in group 5A of the periodic table and has five valence electrons. It thus needs to share three more electrons to make an octet and therefore bonds to three hydrogen atoms, giving PH₃.

✓ Worked Example 2.1.2: Drawing Electron-Dot and Line-Bond Structures

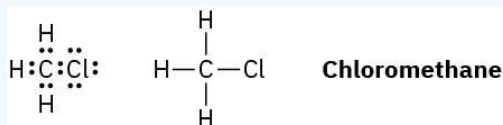
Draw both electron-dot and line-bond structures for chloromethane, CH_3Cl .

Strategy

Remember that a covalent bond—that is, a pair of shared electrons—is represented as a line between atoms.

Solution

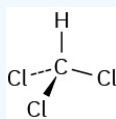
Hydrogen has one valence electron, carbon has four valence electrons, and chlorine has seven valence electrons. Thus, chloromethane is represented as



? Exercise 2.1.1

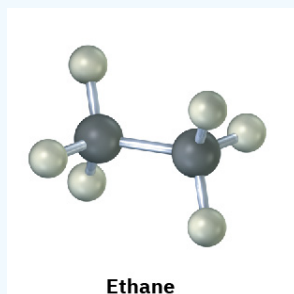
Draw a molecule of chloroform, CHCl_3 , using solid, wedged, and dashed lines to show its tetrahedral geometry.

Answer

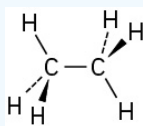


? Exercise 2.1.2

Convert the following representation of ethane, C_2H_6 , into a conventional drawing that uses solid, wedged, and dashed lines to indicate tetrahedral geometry around each carbon (black = C, gray = H).



Answer



? Exercise 2.1.3

What are likely formulas for the following substances?

(a) CCl_4 (b) AlH_3 (c) CH_2Cl_2 (d) SiF_4 (e) CH_3NH_2

Answer

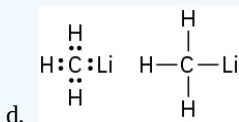
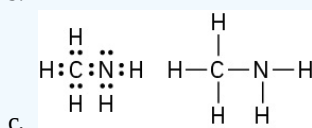
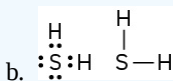
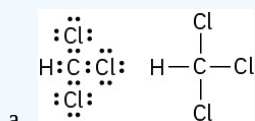
(a) CCl_4 (b) AlH_3 (c) CH_2Cl_2 (d) SiF_4 (e) CH_3NH_2

? Exercise 2.1.4

Write line-bond structures for the following substances, showing all nonbonding electrons:

- CHCl_3 , chloroform
- H_2S , hydrogen sulfide
- CH_3NH_2 , methylamine
- CH_3Li , methyllithium

Answer



? Exercise 2.1.5

Why can't an organic molecule have the formula C_2H_7 ?

Answer

C_2H_7 has too many hydrogens for a compound with two carbons.

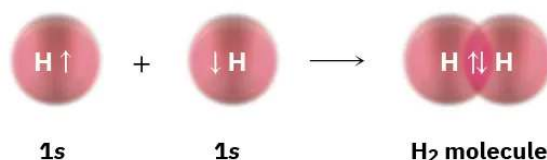
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2.2: Describing Chemical Bonds - Valence Bond Theory

How does electron sharing lead to bonding between atoms? Two models have been developed to describe covalent bonding: *valence bond theory* and *molecular orbital theory*. Each model has its strengths and weaknesses, and chemists tend to use them interchangeably depending on the circumstances. Valence bond theory is the more easily visualized of the two, so most of the descriptions we'll use in this book derive from that approach.

According to valence bond (VB) theory, a covalent bond forms when two atoms approach each other closely and a singly occupied orbital on one atom *overlaps* a singly occupied orbital on the other atom. The electrons are now paired in the overlapping orbitals and are attracted to the nuclei of both atoms, thus bonding the atoms together. In the H_2 molecule, for instance, the H–H bond results from the overlap of two singly occupied hydrogen 1s orbitals.



The overlapping orbitals in the H_2 molecule have the elongated egg shape we might get by pressing two spheres together. If a plane were to pass through the middle of the bond, the intersection of the plane and the overlapping orbitals would be a circle. In other words, the H–H bond is cylindrically symmetrical, as shown in Figure 2.2.1. Such bonds, which are formed by the head-on overlap of two atomic orbitals along a line drawn between the nuclei, are called sigma (σ) bonds.

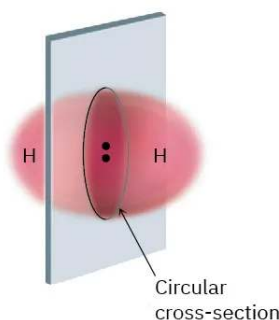


Figure 2.2.1: The cylindrical symmetry of the H–H σ bond in an H_2 molecule. The intersection of a plane cutting through the σ bond is a circle.

During the bond-forming reaction $2H\cdot \rightarrow H_2$, 436 kJ/mol (104 kcal/mol) of energy is released. Because the product H_2 molecule has 436 kJ/mol less energy than the starting 2 $H\cdot$ atoms, the product is more stable than the reactant and we say that the H–H bond has a bond strength of 436 kJ/mol. In other words, we would have to put 436 kJ/mol of energy into the H–H bond to break the H_2 molecule apart into two H atoms (Figure 2.2.2). For convenience, we'll generally give energies in both kilocalories (kcal) and the SI unit kilojoules (kJ): 1 kJ = 0.2390 kcal; 1 kcal = 4.184 kJ.

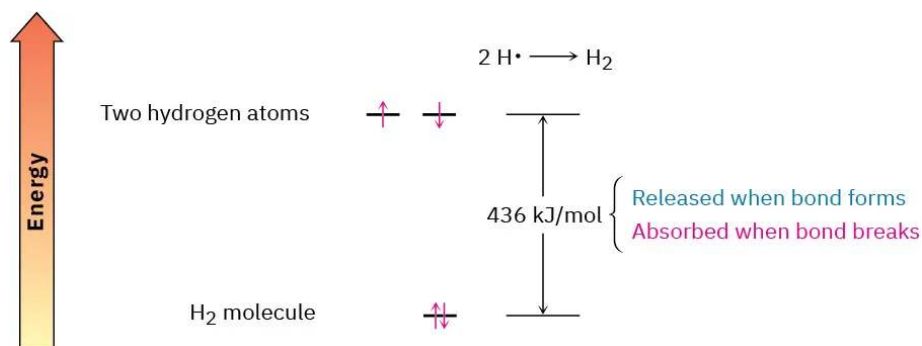


Figure 2.2.2: Relative energy levels of two H atoms and the H_2 molecule. The H_2 molecule has 436 kJ/mol (104 kcal/mol) less energy than the two separate H atoms, so 436 kJ/mol of energy is **released when the H–H bond forms**. Conversely, 436 kJ/mol is **absorbed when the H–H bond breaks**.

How close are the two nuclei in the H_2 molecule? If they are too close, they will repel each other because both are positively charged. Yet if they're too far apart, they won't be able to share the bonding electrons. Thus, there is an optimum distance between nuclei that leads to maximum stability (Figure 2.2.3). Called the bond length, this distance is 74 pm in the $\text{H}-\text{H}$ molecule. Every covalent bond has both a characteristic bond strength and bond length.

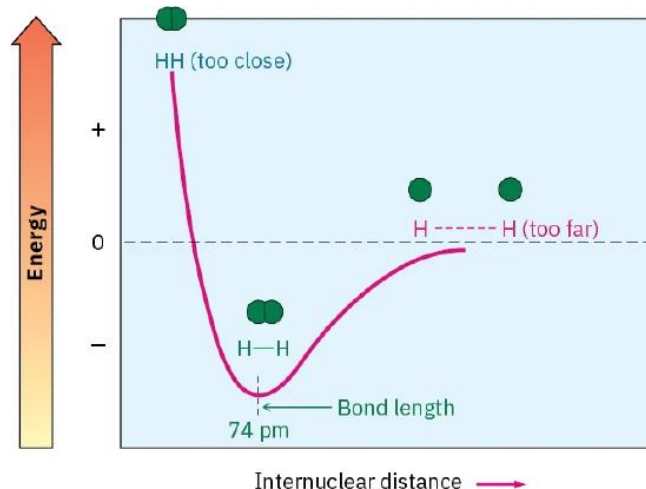


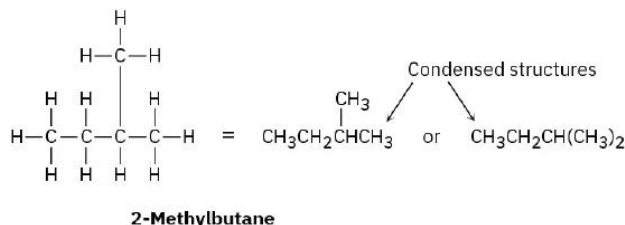
Figure 2.2.3: A plot of energy versus internuclear distance for two hydrogen atoms. The distance between nuclei at the minimum energy point is the **bond length**.

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2.3: Drawing Chemical Structures

Let's cover just one more point before ending this introductory chapter. In the structures we've been drawing until now, a line between atoms has represented the two electrons in a covalent bond. Drawing every bond and every atom is tedious, however, so chemists have devised several shorthand ways for writing structures. In condensed structures, carbon–hydrogen and carbon–carbon single bonds aren't shown; instead, they're understood. If a carbon has three hydrogens bonded to it, we write CH_3 ; if a carbon has two hydrogens bonded to it, we write CH_2 ; and so on. The compound called 2-methylbutane, for example, is written as follows:



Note that the horizontal bonds between carbons aren't shown in condensed structures—the CH_3 , CH_2 , and CH units are simply placed next to each other—but vertical carbon–carbon bonds like that of the first of the condensed structures drawn above is shown for clarity. Notice also in the second of the condensed structures that the two CH_3 units attached to the CH carbon are grouped together as $(\text{CH}_3)_2$.

Even simpler than condensed structures are skeletal structures such as those shown in Table 2.3.1. The rules for drawing skeletal structures are straightforward.

RULE 1

Carbon atoms aren't usually shown. Instead, a carbon atom is assumed to be at each intersection of two lines (bonds) and at the end of each line. Occasionally, a carbon atom might be indicated for emphasis or clarity.

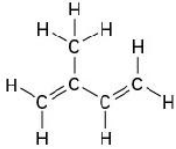
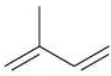
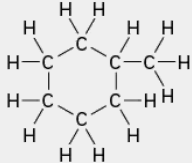
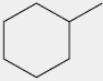
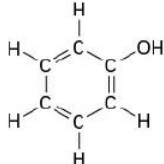
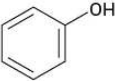
RULE 2

Hydrogen atoms bonded to carbon aren't shown. Because carbon always has a valence of 4, we mentally supply the correct number of hydrogen atoms for each carbon.

RULE 3

Atoms other than carbon and hydrogen *are* shown.

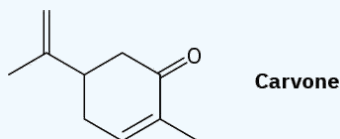
Table 2.3.1 Line-bond and Skeletal Structures for Some Compounds

Compound	Line-bond structure	Skeletal structure
Isoprene, C_5H_8		
Methylcyclohexane, C_7H_{14}		
Phenol, $\text{C}_6\text{H}_6\text{O}$		

One further comment: Although such groupings as -CH_3 , -OH , and -NH_2 are usually written with the C, O, or N atom first and the H atom second, the order of writing is sometimes inverted to $\text{H}_3\text{C-}$, HO- , and $\text{H}_2\text{N-}$ if needed to make the bonding connections clearer. Larger units such as $\text{-CH}_2\text{CH}_3$ are not inverted, though; we don't write $\text{H}_3\text{CH}_2\text{C-}$ because it would be confusing. There are, however, no well-defined rules that cover all cases; it's largely a matter of preference.

✓ Worked Example 2.3.1: Interpreting a Line-Bond Structure

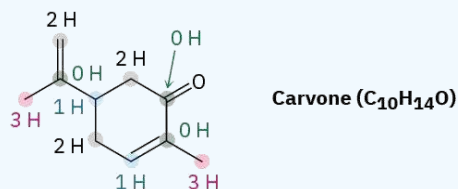
Carvone, a substance responsible for the odor of spearmint, has the following structure. Tell how many hydrogens are bonded to each carbon, and give the molecular formula of carvone.



Strategy

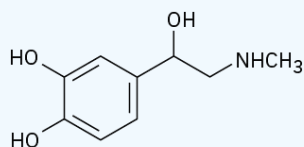
The end of a line represents a carbon atom with 3 hydrogens, CH_3 ; a two-way intersection is a carbon atom with 2 hydrogens, CH_2 ; a three-way intersection is a carbon atom with 1 hydrogen, CH ; and a four-way intersection is a carbon atom with no attached hydrogens.

Solution

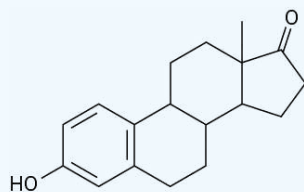


? Exercise 2.3.1

How many hydrogens are bonded to each carbon in the following compounds, and what is the molecular formula of each substance?

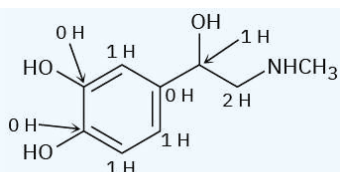


a. **Adrenaline**

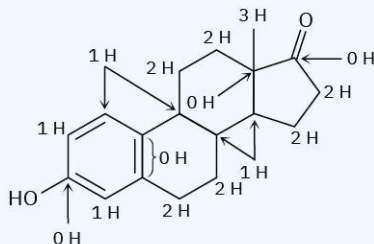


b. **Estrone (a hormone)**

Answer



a. Adrenaline— $C_9H_{13}NO_3$



b. Estrone— $C_{18}H_{22}O_2$

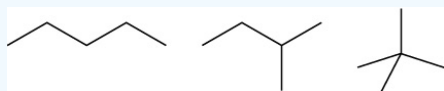
? Exercise 2.3.2

Propose skeletal structures for compounds that satisfy the following molecular formulas: There is more than one possibility in each case.

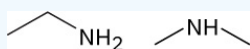
- C_5H_{12}
- C_2H_7N
- C_3H_6O
- C_4H_9Cl

Answer

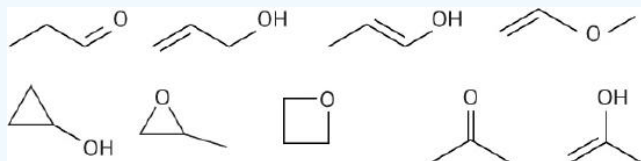
(a) There are numerous possibilities, such as:



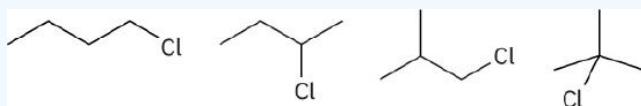
(b) There are numerous possibilities, such as:



(c) There are numerous possibilities, such as:

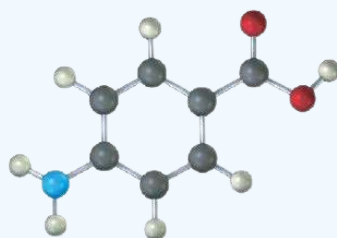


(d) There are numerous possibilities, such as:

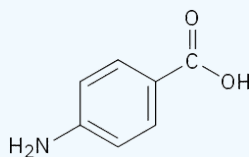


? Exercise 2.3.3

The following molecular model is a representation of *para*-aminobenzoic acid (PABA), the active ingredient in many sunscreens. Indicate the positions of the multiple bonds, and draw a skeletal structure (black = C, red = O, blue = N, gray = H).



Answer



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2.4: sp^3 Hybrid Orbitals and the Structure of Methane

The bonding in the hydrogen molecule is fairly straightforward, but the situation is more complicated in organic molecules with tetravalent carbon atoms. Take methane, CH_4 , for instance. As we've seen, carbon has four valence electrons ($2s^2 2p^2$) and forms four bonds. Because carbon uses two kinds of orbitals for bonding, $2s$ and $2p$, we might expect methane to have two kinds of C–H bonds. In fact, though, all four C–H bonds in methane are identical and are spatially oriented toward the corners of a regular tetrahedron. How can we explain this?

An answer was provided in 1931 by Linus Pauling, who showed mathematically how an s orbital and three p orbitals on an atom can combine, or *hybridize*, to form four equivalent atomic orbitals with tetrahedral orientation. Shown in Figure 2.4.1, these tetrahedrally oriented orbitals are called sp^3 hybrid orbitals. Note that the superscript 3 in the name sp^3 tells how many of each type of atomic orbital combine to form the hybrid, not how many electrons occupy it.

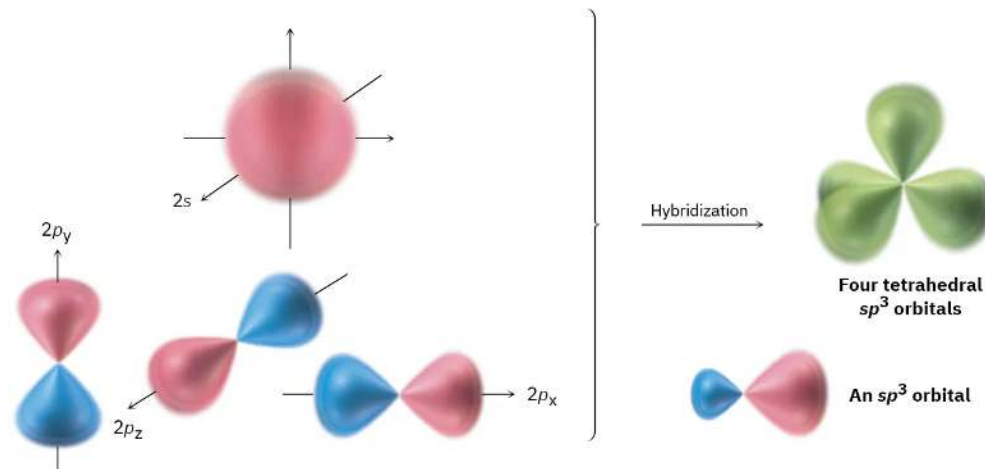


Figure 2.4.1: Four sp^3 hybrid orbitals, oriented toward the corners of a regular tetrahedron, are formed by the combination of an s orbital and three p orbitals (red/blue). The sp^3 hybrids have two lobes and are unsymmetrical about the nucleus, giving them a directionality and allowing them to form strong bonds to other atoms.

The concept of hybridization explains how carbon forms four equivalent tetrahedral bonds but not why it does so. The shape of the hybrid orbital suggests the answer to why. When an s orbital hybridizes with three p orbitals, the resultant sp^3 hybrid orbitals are unsymmetrical about the nucleus. One of the two lobes is larger than the other and can therefore overlap more effectively with an orbital from another atom to form a bond. As a result, sp^3 hybrid orbitals form stronger bonds than do unhybridized s or p orbitals.

The asymmetry of sp^3 orbitals arises because, as noted previously, the two lobes of a p orbital have different algebraic signs, $+$ and $-$, in the wave function. Thus, when a p orbital hybridizes with an s orbital, the positive p lobe adds to the s orbital but the negative p lobe subtracts from the s orbital. The resultant hybrid orbital is therefore unsymmetrical about the nucleus and is strongly oriented in one direction.

When each of the four identical sp^3 hybrid orbitals of a carbon atom overlaps with the $1s$ orbital of a hydrogen atom, four identical C–H bonds are formed and methane results. Each C–H bond in methane has a strength of 439 kJ/mol (105 kcal/mol) and a length of 109 pm. Because the four bonds have a specific geometry, we also can define a property called the bond angle. The angle formed by each H–C–H is 109.5° , the so-called tetrahedral angle. Methane thus has the structure shown in Figure 2.4.2.

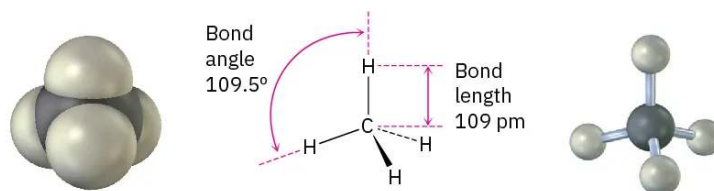


Figure 2.4.2: The structure of methane, showing its 109.5° bond angles.

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